

Package ‘multChernoff’

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Title Finite-Sample Tail Bounds of Likelihood Ratio Test under Multinomial Sampling
Version 0.1.0
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Description Computes finite-sample tail bounds for the likelihood ratio test (LRT) under multinomial sampling. The resulting bounds can be used to form conservative p-values and critical values when standard chi-squared asymptotics may be inaccurate.
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Encoding UTF-8
Roxygen list(markdown = TRUE)
RoxygenNote 7.3.3
URL <https://github.com/richardkwo/multChernoff>
Imports plyr
Suggests knitr, rmarkdown, testthat (>= 3.0.0)
VignetteBuilder knitr
Config/testthat/edition 3
NeedsCompilation no

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criticalValue	<i>Critical value x such that $P(LRT > x) \leq p$</i>
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Description

Critical value x such that $P(LRT > x) \leq p$

Usage

```
criticalValue(k, n, p = 0.05, verbose = FALSE)
```

Arguments

k	number of categories (a vector for independent multinomial draws)
n	sample size (a vector for independent multinomial draws)
p	significance level (e.g., 0.05)
verbose	draw the minimizer if TRUE

See Also

tailProbBound

Examples

```
n <- 1:40
crit <- sapply(n, function(.n) criticalValue(20, .n, p=0.01))
plot(n, crit)
abline(h=qchisq(0.01, df=20-1, lower.tail = FALSE)) # standard chi-square asymptotic
criticalValue(10, 40, p=0.05)
```

mgfBound	<i>$G(\lambda; k, n)$ upper bound of the moment generating function on $(LRT / 2)$</i>
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Description

$G(\lambda; k, n)$ upper bound of the moment generating function on $(LRT / 2)$

Usage

```
mgfBound(k, n, lambda)
```

Arguments

k	number of categories
n	sample size
lambda	number between 0 and 1

tailProbBound	<i>Tail bound on $P(LRT > x)$</i>
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Description

Tail bound on $P(LRT > x)$

Usage

```
tailProbBound(x, k, n, verbose = FALSE)
```

Arguments

x	the value of the LRT
k	number of categories (a vector for independent multinomial draws)
n	sample size (a vector for independent multinomial draws)
verbose	draw the minimizer if TRUE

Value

an upper bound on $P(LRT > x)$, which can be used as a conservative p-value

Note

For independent multinomial samples, k and n must be of the same length.

Examples

```
tailProbBound(20, 7, 50)
pchisq(20, 6, lower.tail=FALSE) # compare with the standard chi-square asymptotic
tailProbBound(12, c(3, 4), c(12, 20))
pchisq(12, 5, lower.tail=FALSE) # compare with the standard chi-square asymptotic
```